

TEAM DOCUMENTATION REPORT

RARITONE BACKEND SYSTEM

BACKEND TEAM

1. Introduction

The Raritone Backend System was developed as a collaborative team project during the Raritone Summer Internship 2026. The primary objective of the project was to build a secure, scalable, modular, and AI-ready backend infrastructure for a virtual fashion platform. The system was designed to support user management, product management, shopping workflows, virtual try-on preparation, authentication, authorization, media management, and future AI-driven functionalities.

The project was developed using modern backend technologies including Node.js, Express.js, MongoDB, JWT Authentication, Cloudinary, and various security mechanisms. Team members collaborated across multiple modules to ensure smooth integration, maintainability, and future scalability of the platform.

This report summarizes the team structure, individual responsibilities, collective contributions, and overall project outcomes achieved during the internship period.

2. Team Structure

The backend development team consisted of six members who worked collaboratively on different modules and functionalities of the system.

Team Members:

- Nandini Bhimineni – Team Lead
- Vagdevi Malineni
- Bharath Kumar
- Vishnu Vardhan Reddy
- Meka Hrishika Teja Chowdary
- Nainisha Bandari

3. Individual Contributions

Nandini Bhimineni – Team Lead

As Team Lead, Nandini coordinated development activities, module integration, documentation, testing, and future planning initiatives. She played a significant role in ensuring that different modules developed by team members were successfully integrated into a unified backend system.

Responsibilities:

- Backend Integration
- Cart Module Development
- Order Module Development
- Measurement Module Development
- Avatar Module Development
- Product Mapping Module Development
- Database Relationship Design
- API Testing and Verification
- Cloud Infrastructure Research
- Scalability Planning
- Documentation Coordination

Key Contributions:

- Coordinated backend integration across modules.
- Developed shopping cart and order management workflows.
- Designed measurement, avatar, and product mapping systems.
- Conducted cloud infrastructure and scalability research.
- Led project documentation and reporting activities.
- Performed API testing and validation before deployment.

Vagdevi Malineni

Vagdevi primarily contributed to authentication and security-related modules. Her work focused on secure user access management and profile-related functionalities.

Responsibilities:

- Authentication Module
- JWT Security Implementation
- Role-Based Access Control (RBAC)
- Profile Module
- Wardrobe Module
- Protected Routes
- Security Features

Key Contributions:

- Implemented authentication workflows.
- Developed profile and wardrobe management features.
- Integrated role-based access controls.
- Developed protected routes and security mechanisms.
- Enhanced overall backend security architecture.

Bharath Kumar

Bharath contributed to authentication support, wishlist functionality, validation systems, and testing activities.

Responsibilities:

- Authentication Module Support
- Wishlist Module
- API Validation
- Security Enhancements
- Avatar Module Support
- Backend Testing Support

Key Contributions:

- Assisted in authentication implementation.
- Developed wishlist management functionality.
- Contributed to API validation and error handling.
- Supported avatar module development.
- Participated in testing and debugging activities.

Vishnu Vardhan Reddy

Vishnu contributed to wishlist development, virtual try-on workflows, security implementation, and testing support.

Responsibilities:

- Wishlist Module
- Virtual Try-On Module
- Authentication Support
- Security Implementation
- API Testing
- Workflow Validation

Key Contributions:

- Developed wishlist functionalities.
- Assisted in virtual try-on workflow development.
- Supported authentication and security implementation.
- Performed API testing and validation.
- Verified backend workflow execution and integration.

Meka Hrishi Teja Chowdary

Hrishi primarily contributed to product management, product APIs, order processing workflows, and virtual try-on module development.

Responsibilities:

- Product Module Development
- Product API Development
- Product Management System
- Order Module Contribution
- Virtual Try-On Module Development
- Try-On Workflow Design

Key Contributions:

- Developed product management functionalities.
- Designed and implemented product APIs.
- Contributed to order management workflows.
- Assisted in virtual try-on module development.
- Participated in workflow planning and implementation.

Nainisha Bandari

Nainisha contributed to user profile management and wardrobe-related functionalities.

Responsibilities:

- Profile Module Development
- Wardrobe Module Development
- User Profile Management
- User Preference Management
- Profile API Development

Key Contributions:

- Developed profile management features.
- Implemented user preference management functionality.
- Assisted in wardrobe module development.
- Developed profile-related APIs.
- Supported user-centric backend operations.

4. Collective Team Contributions

The team collaborated extensively across multiple technical and research domains to ensure successful project completion.

Backend Development:

- REST API Development
- Backend Architecture Design
- Controller Development
- Route Management
- Modular System Development

Database Development:

- MongoDB Schema Design
- Collection Management
- Database Relationships
- ObjectId Referencing
- Data Management Strategies

Authentication and Security:

- JWT Authentication
- Google OAuth Integration
- Role-Based Access Control (RBAC)
- Protected Routes
- Input Validation
- Security Middleware Integration

Cloud and Storage:

- Cloudinary Integration
- Image Upload Management
- Media Storage Research
- Cloud Infrastructure Planning

Testing and Quality Assurance:

- API Testing using Postman
- Endpoint Validation
- Workflow Testing
- Integration Testing
- Error Handling Verification

Documentation:

- Technical Documentation
- API Documentation
- Project Reports
- Presentation Preparation
- Knowledge Sharing Sessions

Future Planning:

- AI Integration Planning
- Virtual Try-On Architecture Preparation
- Scalability Research
- Cloud Infrastructure Research
- Future Technology Evaluation

5. Project Outcomes

The combined efforts of the team resulted in the successful development of a robust backend foundation for the Raritone platform.

Major Outcomes:

- Secure Backend Infrastructure
- Modular Backend Architecture
- RESTful API Ecosystem
- Authentication and Authorization System
- Product Management System
- Order Processing Workflows
- User Profile and Wardrobe Management
- Avatar and Virtual Try-On Preparation
- Database Relationship Implementation
- Cloud Integration Research
- Scalability Planning
- AI-Ready Backend Foundation

6. Skills Developed Through Team Collaboration

Throughout the project, team members gained practical experience in:

- Backend Development
- API Design and Development
- MongoDB Database Management
- Authentication and Security
- Testing and Debugging
- Documentation Practices
- Version Control using GitHub
- Team Communication
- Project Planning and Coordination
- Problem Solving and Research

7. Conclusion

The Raritone Backend System was successfully developed through effective teamwork, collaboration, and shared technical responsibilities. Each team member contributed to specific modules while working collectively toward the successful completion of the project.

The internship provided valuable industry exposure in backend development, database design, API creation, authentication systems, testing procedures, documentation, and collaborative software engineering practices. The resulting backend platform serves as a strong foundation for future enhancements, scalability initiatives, AI integration, and virtual fashion technologies.

